

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418190
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Montambault; Serge et al.

System for cycling battery cells with pressure regulation

Abstract

A system for cycling battery cells with pressure regulation is disclosed. The system comprises supports for receiving the battery cells which insert in a clamping arrangement having jaws moving one with respect to the other by an actuator operated by a controller in order to apply a pressure in a pressure application axis to the cells. A pressure sensor measures the pressure applied to the cells while a cycling module connected to the cells performs their cycling and measures their charge and discharge level. A programmable processing unit ensures a control of the pressure and, if desired, the temperature applied to the cells according to a programmed cycling mode and records data representative of the pressure and other cycling measurements.

Inventors:	Montambault; Serge (Saint-Marc-sur-Richelieu, CA), Morin; François (Carignan, CA), Rivard; Étienne (Boucherville, CA), Perreault; Christian (Saint-Bruno-de-Montarville, CA), Zaghib; Karim (Longueuil, CA)
Applicant:	HYDRO-QUÉBEC (Montréal, CA)
Family ID:	1000008819416
Assignee:	HYDRO-QUÉBEC (Montreal, CA)
Appl. No.:	17/998579
Filed (or PCT Filed):	May 07, 2021
PCT No.:	PCT/CA2021/050642
PCT Pub. No.:	WO2021/226705
PCT Pub. Date:	November 18, 2021

Prior Publication Data

Document Identifier	Publication Date
---------------------	------------------

Foreign Application Priority Data

CA

CA 3080727

May. 13, 2020

Publication Classification**Int. Cl.:** H02J7/00 (20060101)**U.S. Cl.:**

CPC H02J7/0069 (20200101); H02J7/0013 (20130101); H02J7/0047 (20130101);

Field of Classification Search**CPC:** H01M (10/486)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4209479	12/1979	Gunther et al.	N/A	N/A
7264901	12/2006	Gow et al.	N/A	N/A
7591303	12/2008	Zeigler	62/323.3	B60H 1/00378
8440339	12/2012	Harada et al.	N/A	N/A
8507125	12/2012	Meschter	N/A	N/A
9306252	12/2015	Kristofek et al.	N/A	N/A
9634358	12/2016	Matsushita	N/A	N/A
2009/0087727	12/2008	Harada	429/120	H01M 10/613
2009/0123819	12/2008	Kim	429/120	F28D 15/0275
2010/0276132	12/2009	Payne	165/169	F28D 15/0266
2010/0304203	12/2009	Buck	29/730	H01M 10/647
2011/0027631	12/2010	Koenigsmann	429/62	H01M 10/6567
2011/0052959	12/2010	Koetting	429/120	H01M 10/6555
2011/0293982	12/2010	Martz	429/120	H01M 10/6557
2012/0244393	12/2011	Stanek et al.	N/A	N/A
2013/0255293	12/2012	Gadawski	429/62	H01M 10/6555
2013/0280575	12/2012	Obrist et al.	N/A	N/A
2013/0330577	12/2012	Kristofek et al.	N/A	N/A

2014/0147709	12/2013	Ketkar	429/50	F25B 1/00
2015/0134172	12/2014	Ose	N/A	N/A
2015/0303527	12/2014	Maxwell	N/A	N/A
2016/0181674	12/2015	Kristofek et al.	N/A	N/A
2016/0285144	12/2015	Song et al.	N/A	N/A
2017/0125860	12/2016	Chatroux et al.	N/A	N/A
2018/0191023	12/2017	Cao	N/A	N/A
2019/0245170	12/2018	Brinkmann et al.	N/A	N/A
2023/0049302	12/2022	Kim	N/A	H01M 10/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2306482	12/1998	CA	N/A
107732310	12/2017	CN	H01M 10/0404
107256934	12/2018	CN	N/A
102013213909	12/2014	DE	N/A
2924797	12/2016	EP	N/A
2010040345	12/2009	JP	N/A
5266634	12/2012	JP	N/A
2014010983	12/2013	JP	N/A
2020119823	12/2019	JP	N/A
2014178236	12/2013	WO	N/A
2019123903	12/2018	WO	N/A
WO-2020239927	12/2019	WO	H02J 7/0069

OTHER PUBLICATIONS

International Search Report and Written Opinion for international application No. PCT/CA2021/050642, mailed on Aug. 25, 2021. cited by applicant

Primary Examiner: Kessie; Daniel

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase Under 35 U.S.C.  of International Application No. PCT/CA2021/050642, titled “SYSTEM FOR CYCLING BATTERY CELLS WITH PRESSURE REGULATION,” filed May 7, 2021, which is a PCT application claiming priority to and the benefit of Canadian Application No. 3,080,727, titled “SYSTEM FOR CYCLING BATTERY CELLS WITH PRESSURE REGULATION,” filed May 13, 2020, which are incorporated herein in their entirety by reference.

FIELD OF THE INVENTION

(2) The invention generally relates to equipment for cycling battery cells, and more particularly to a system for cycling battery cells with pressure regulation.

BACKGROUND

(3) New batteries are constantly being developed, in particular to increase the range of an electric vehicle. In addition to batteries with a liquid electrolyte such as lithium-ion, searches also focus on solid-state batteries as well as hybrid-type batteries (liquid and solid electrolyte). Solid-state batteries are increasingly considered to advantageously replace the liquid electrolyte batteries and form the next generations of batteries, especially in the electric vehicle industry and residential and industrial energy storage. The teams working on battery chemistry need systems for cycling cells which they develop to arrive at an efficient product. Among other things, the chemistry and the architecture of solid-state battery cells involve that they are subjected to a precise pressure to get optimum performance of the battery. It is thus important to determine the pressure value to be applied to the cells to obtain optimum performance. In the case of a solid-state battery, the optimum pressure may be significant, for example several hundreds of PSI and even over 4000 PSI, even if a minimum pressure is desired.

(4) The current systems for cycling under pressure are inappropriate for determining the parameters optimizing the performance of a battery, depending on the cell chemistry used. Although the current systems may regulate the temperature using a device with a heat-transfer liquid, they only allow a moderate pressure application on the cells, generally of some PSI and not exceeding about 100 PSI. The current systems thus do not allow applying significant levels of pressure as required for solid-state batteries, do not allow applying a programmable variable pressure, and do not allow controlling the pressure application function. Also, the current systems are complex to adjust, heavy, or have little flexibility. Furthermore, the current systems have temperature regulation circuits configured to maintain the test temperature of a battery at a fixed value, for example 27° C. (80° F.) for a Li-ion type battery, whereas a variable temperature could be preferable depending on the chemistry of the battery.

SUMMARY

(5) According to an embodiment of the invention, there is provided a system for cycling battery cells, the system comprising: at least one support defining a housing for receiving one of the battery cells; a clamping arrangement having (i) lower and upper jaws between which said at least one support housing one of the battery cells is insertable, the jaws being mobile with respect to each other in a pressure application axis, and (ii) pressure application surfaces for applying pressure to the cell housed in each support, one of the pressure application surfaces being intended to exert a pressure on an active zone of the cell when the jaws are in pressure application position; an actuator operatively coupled to the clamp arrangement in order to control a clamping of the jaws according to a control signal; a pressure sensor operatively associated to the cell in order to produce a signal indicative of a pressure exerted by the jaws on the cell; a cycling module for cycling the cell, the cycling module having an interface connectable to the cell for charging and discharging the cell according to a programmed cycling mode and measuring a level of charge and discharge of the cell; a control module for controlling the actuator, producing the control signal controlling the clamping of the jaws according to a pressure setting signal; and a processing unit connecting to the cycling module, to the control module of the actuator and to the pressure sensor, the processing unit being configured to program the cycling mode of the cell and transmit the cycling mode to the cycling module, to program a pressure to be applied to the cell and generate the pressure setting signal according to a pressure measured by the pressure sensor in order to form a feedback loop of the pressure applied to the cell, and to record data representative of the pressure applied to the cell and of the level of charge and discharge of the cell.

(6) Preferably, the system according to the invention may be used as a tool for determining one or several precise values of pressures to be applied to battery cells, as well as other possible parameters influencing their behaviors and operations such as one or more precise temperatures, to obtain optimum performance of the cells according to their chemistries and various conditions of operation and use tested by the system, including a rate of charge and discharge and a density of transited current. The system allows cycling the cells to be tested under low to high pressures and

developing a real-time management algorithm implementable for example in a BMS (“Battery Management System”) or a VCU (“Vehicle Control Unit”) to control a pressure to be applied and a temperature to be maintained or varied for optimizing a performance of the battery as a function of various conditions of operation and use of the cells.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A detailed description of preferred embodiments of the invention will be given hereinbelow with reference to the following drawings:

(2) FIG. 1 is a block diagram illustrating a battery cycling system according to an embodiment of the invention.

(3) FIG. 2 is a perspective and exploded view of a pressure application device for applying pressure to the cells according to an embodiment of the invention.

(4) FIG. 3 is a perspective view of a pressure application device with cell supports prior to their positioning in the device, according to an embodiment of the invention.

(5) FIG. 4 is a perspective view from another angle of the device and the cell supports illustrated in FIG. 3.

(6) FIG. 5 is a perspective and partially exploded view of a pressure application device with cell supports during their insertion in the device, according to an embodiment of the invention.

(7) FIG. 6 is a perspective and exploded view of a part of a pressure application device according to an embodiment of the invention.

(8) FIG. 7 is a perspective view of a pressure application device in which cell supports are inserted, according to an embodiment of the invention.

(9) FIG. 8 is a perspective view for another angle of the pressure application device illustrated in FIG. 7.

(10) FIG. 9 is a perspective and exploded view of a cell support according to an embodiment of the invention.

(11) FIG. 10 is a perspective view of the cell support according to an embodiment of the invention.

(12) FIGS. 11A, 11B and 11C respectively illustrate examples of cycling protocols of a battery cell according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) Referring to FIG. 1, a system for cycling battery cells 2 according to an embodiment of the invention is illustrated.

(14) Referring to FIGS. 2 to 8, the system comprises one or more supports 4 defining a housing 6 (as illustrated in FIG. 9) for receiving one of the battery cells 2. To simplify the following description, reference will generally be made to several supports 4 and several cells 2 but it should be understood that there may be only one support 4 and only one cell 2 as in the arrangement illustrated without support in FIG. 6.

(15) Referring in particular to FIG. 2, a clamping arrangement 8 has lower and upper jaws 10, 12 between which the supports 4 housing battery cells 2 are insertable. The jaws 10, 12 are mobile with respect to each other in a pressure application axis as depicted by arrow 14. The clamping arrangement 8 also has pressure application surfaces 16, 18 for application of pressure to the cell 2 housed in each support 4. According to an embodiment, the pressure application surfaces 18 are intended to exert a pressure on active zones 20 (e.g. anode-cathode-electrolyte complex, as illustrated in FIG. 9) of the cells 2 when the jaws 10, 12 are in pressure application position. A reverse embodiment where the surfaces 16 are intended to exert the pressure on the active zones 20 is also possible. The pressure application surfaces 16, 18 may be formed by surfaces of the jaws 10, 12 and surfaces of pressure transmission plates 42 extending between two supports 4, each pressure

transmission plate **42** being freely mobile in the pressure application axis **14**.

(16) An actuator **22** is operatively coupled to the clamping arrangement **8** so as to control a clamping of the jaws **10, 12** according to a control signal **24** (as illustrated in FIG. 1).

(17) A pressure sensor **26** is operatively associated to the cells **2** in order to produce a signal **28** (as illustrated in FIG. 1) indicative of a pressure exerted by the jaws **10, 12** on the cells **2**.

(18) Referring back to FIG. 1, the system according to the invention comprises a cycling module **30** for cycling the cells **2**, also called battery cycler. The cycling module **30** has an interface **32** connectable to the cells **2** for charging and discharging the cells **2** according to a programmed test cycling mode or cycling protocol and for measuring levels of charge and discharge of the cells **2**, for example by voltage and current measurements relative to the cells **2**. In charging mode, the cycling module **30** injects a current having an intensity depending on a desired charging speed. This intensity may be qualified in C. A value of 1C means that a cell **2** will be charged in 1 hour, a value of 2C means that the cell **2** will be charged in 30 minutes, a value of C/2 means that the cell **2** will be charged in 2 hours, etc. Likewise, in discharging mode, the cycling module **30** consumes the energy stored in a cell **2** and extracts a current therefrom having an intensity depending on a desired discharging speed which may also be measured in C. A control module **34** for controlling the actuator **22**, also called motor controller, produces the control signal **24** that controls the clamping of the jaws **10, 12** (illustrated e.g. in FIG. 2) according to a pressure setting signal **36**. A processing unit **38** connects to the cycling module **30**, to the control module **34** of the actuator **22** and to the pressure sensor **26**. The processing unit **38** is configured to program the cycling mode of the cells **2** and transmit it to the cycling module **30**, to program a pressure to be applied to the cells **2**, for example as a function of a level or a rate of charge and discharge or as a function of a number of cycles imposed to the cells **2**, and to generate the pressure setting signal **36** according to a pressure measured by the pressure sensor **26** so as to form a feedback loop of the pressure applied to the cells **2**, and to record data representative of the pressure applied to the cells **2** and the levels of charge and discharge of the cells **2**. The processing unit **38** has an interface **40** for communicating with the different components of the system and with the outside, in order for example to set parameters of the functions of the cycling system and transmit the data recorded by the processing unit **38**. The processing unit **38** allows modulating the pressure applied to the cells **2** as a function, for example, of two parameters managed by the cycling module **30**, namely the level of charge/discharge of the cells **2**, measured and evaluated by the cycling module **30** as a function of the voltage level of the cells **2**, and the charging rate, measured and evaluated by the cycling module **30** as a function of the intensity of current transited between the cycling module **30** and the cells **2**. As a function of the established test protocol, a feedback on the level of charge/discharge and the rate of charge/discharge is transmitted to the processing unit **38** which coordinates pressure value variation instructions transmitted via the control module **34** of the actuator **22** to adjust the value of pressure applied by the actuator **22** and the clamping arrangement **8**.

(19) Referring to FIG. 11A, an example of a pressure value instruction forming the control signal **24** transmitted by the control module **34** to the actuator **22** is illustrated, as a function of the number of cycles reached by a cell **2** and the value of its charge/discharge level. According to a possible test protocol **104**, a high and constant pressure is applied during a charging period of the cell **2**, while a lower and constant pressure is applied during a discharging period of the cell **2**. According to another possible test protocol **106**, a same constant pressure is applied during the charging and discharging phases of a cell **2** for a same cycle, and then the pressure value is slightly increased from cycle to cycle, as a function for example of an expected decrease of the useful life of the cell **2**. Other test protocols are of course possible.

(20) Referring to FIG. 11B, the processing unit **38** may coordinate the instructions related to a test protocol **108** between the control module **34** of the actuator **22** and the cycling module **30** so as to increase the mean value of the pressure applied to a cell **2** as a function of the increase of the value of the charge/discharge rate measured by the intensity of the current transited between the cycling

module **30** and the cell **2**. By modulating the pressure, the performance of the cell **2** can thus be maximized under greater response conditions, such as current demands involving discharge rates of several C. This pressure modulation may also be used in order to prevent a premature deterioration of the cell **2** during extraordinary operating conditions.

(21) Referring to FIG. **11C**, one of the effects of the application of the pressure on the performance of the battery that the cycling system according to the invention allows evaluating is illustrated. For instance, a test protocol **110** allows achieving a higher number of cycles of charge and discharge of a cell **2** with respect to a test protocol **112** where a lower value of pressure is applied to the cell **2**, thus reducing more quickly the useful life of the cell **2**. Other effects on the performance of the cell **2** may be observed by varying the values of pressure and, if desired, of temperature during charging and discharging, such as an intensity of admissible charge/discharge current without damaging the cell **2**, a maintenance of the capacity of the cell **2** during cycling, performance of the cell **2** at low temperature, a contained propagation of dendrites, etc.

(22) Referring back to FIG. **2**, according to an embodiment, the clamping arrangement **8** comprises a set of guiding rods **44** extending between the jaws **10**, **12** in the pressure application axis **14**. The guiding rods **44** pass through corresponding openings **45** in each pressure transmission plate **42** so that each pressure transmission plate **42** is mobile only in the pressure application axis **14**.

(23) According to an embodiment, the upper jaw **12** is formed by a pressure transmission plate freely mobile in the pressure application axis **14**, such as the pressure transmission plates **42**. The actuator **22** comprises a motor unit **46**, a piston **48** coupled to and moveable by the motor unit **46** in the pressure application axis **14** according to the control signal **24** (illustrated in FIG. **1**) transmitted to the motor unit **46**. The actuator **22** may be of a linear type, and be embodied by a hydraulic or pneumatic jack or, preferably, by an electric jack possibly more precise, more compact, easier to control and having a faster response time. The actuator **22** also comprises a support structure **50** for supporting the motor unit **46** at a fixed distance from the lower jaw **10**. The piston **48** has an end **52** pressing against the transmission plate forming the upper jaw **12** when the jaws **10**, **12** are in the pressure application position. The support structure **50** may take the form of a plate **54** connected to the lower jaw **10** by the guiding rods **44**. The motor unit **46** is mounted on the plate **54**. The plate **54** has a central hole **56** through which the piston **48** passes to apply a pressure against the transmission plate forming the upper jaw **12**.

(24) Referring to FIG. **4**, according to an embodiment, each pressure transmission plate **42** has an upper surface **16** defining a positioning cavity **58** in which a lower part of a support **4** is engageable.

(25) According to an embodiment, the clamping arrangement **8** comprises springs **60** extending between the jaws **10**, **12** and each pressure transmission plate **42** so that the jaws **10**, **12** and the pressure transmission plates **42** are moved away from one another by a distance of at least a height of one support **4** housing a cell **2** when the jaws **10**, **12** are in release position, thereby facilitating the insertion of the supports **4** in the clamping arrangement **8**.

(26) Referring again to FIG. **2**, according to an embodiment, the lower jaw **10** has an upper surface **16** defining a cavity **62** having a shape adapted to receive the pressure sensor **26** so that the pressure sensor **26** measures a pressure applied by the jaws **10**, **12** to the cells **2** in the supports **4**. The lower jaw **10** may advantageously serve as or form a base for the clamping arrangement **8**.

(27) According to an embodiment, the pressure application surfaces **16**, **18** are generally flat and extend perpendicular to the pressure application axis **14**.

(28) Referring back to FIG. **1**, the processing unit **38** may comprise a processor **64** coupled to a memory **66** which stores instructions executable by the processor **64** and operating parameters of the cycling system received in input **68** through for example the input/output interface **40**. The operating parameters may comprise a test protocol used by the instructions executed by the processor **64** and defining the pressure to be applied. In the case where a thermal regulation or control is considered, the instructions may define a temperature value of the cells **2** to be reached as

a function for example of a level and a rate of charge and discharge derived from the instructions and measurements associated to the cycling module **30** of the cells **2** in communication with the processing unit **38**. The test protocol may define a series of pressures to be applied and maintained during a time period, for example from 0 PSI to 4000 PSI or even more if desired. The pressure setting signal **36** transmitted to the control module **34** of the actuator **22** may be regulated by the processing unit **38** in compensation to a variation of volume of the cells **2** during an execution of the cycling mode. A thickness, distance or position sensor (not shown) of the cells **2** may be operatively associated to the cells **2** so as to produce a signal indicative of their thickness or a variation of their thickness during the cycling. The signal may be transmitted to the processing unit **38** which may record data representative of the thicknesses measured by the thickness sensor. The measured thicknesses may be used for the regulation of pressure in the feedback loop, such that for example the pressure applied to the cells **2** is maintained fixed even in presence of a variation of temperature or volume of the cells **2**. The instructions executed by the processor **64** may comprise instructions for dynamically varying the series of pressures to be applied and maintained as a function of cycles of charging and discharging of the cells **2**, power demands, a number of cycles achieved, etc. The instructions may include a regulation control algorithm of the actuator **22** via the control module **34** and according to a feedback of the pressure measured by the pressure sensor **26**. The pressure to be applied may also depend on a pressure of oil or like fluid in the case where the actuator **22** is a hydraulic or pneumatic jack, or a current in the case where the motor unit **46** is electric.

(29) A temperature sensor **70** may be operatively associated to the cells **2** in order to produce a signal indicative of a temperature of the cells **2**. The processing unit **38** connects to the temperature sensor **70** and is configured to record data representative of the temperature of the cells **2** according to the signal indicative of the temperature of the cells **2**. The pressure setting signal **36** may then be adjusted according to the temperature measured by the temperature sensor **70**. The system may comprise a temperature adjustment module **96** connected to the processing unit **38** via its interface **40** to act on a heat-exchange element **98** flowable around the cells **2** in the clamping arrangement **8** in order to for example maintain a uniform temperature from one cell **2** to another, as a function of a temperature signal **100** measured by the temperature sensor **70** and a temperature setpoint **102** generated by the processing unit **38**. The action of the heat-exchange element **98** for dynamically controlling the temperature of the cells **2** may take the form of an aeration of the environment of the supports **4** (illustrated e.g. in FIG. 2) for example by disposing the clamping arrangement **8** or a part thereof containing the cells **2** in a heat chamber (not illustrated) allowing a cycling at different temperatures. The values of temperature to be applied during a cycling may be programmed in the processing unit **38** and controlled via the temperature adjustment module **96** during tests on a battery. The action of the heat-exchange element **98** may also be implemented via a network of pipes (not illustrated) running through the supports **4** and in which a heat-exchange liquid flows for controlling the temperature of the cells **2** in a dynamic way in relation for example with the instructions of the test protocol which may define values of pressure or temperature to be applied to the cells **2** during a time period, as a function of or regardless the cycling parameters such as a level and a rate of charge and discharge of the cells **2** and a number of cycles imposed to the cells **2**. The temperature adjustment module **96** may be embodied by a device comprising a pump, tanks, servo valves, a flow meter, a pipe, and a microcontroller configured to respond to the temperature value setpoints of the test protocol and to control the elements of the device accordingly.

(30) The programming of the test protocol may be achieved via a software stored in the memory **66** and executed by the processor **64**, in order for example to display a user-friendly graphical user interface allowing entering values of pressure to be applied, of temperature, etc., which may vary through time as a function of the level of charge/discharge of a cell **2** (voltage), rates of charge/discharge (current intensity), the number of cycles experienced by the battery, the residual capacity of the battery, its variation of thickness, etc. The software may be used to transmit the

setpoints to the control module **34** of the actuator **22**, to the cycling module **30** of the cells **2** and to the temperature adjustment module **96**. The control for reaching the setpoints is achieved via the integration of feedback coming from the pressure sensor **26**, the temperature sensor **70** and the measurements performed via the cycling module **30** (voltage, current, time), by the processing unit **38** via its interface **40** which integrates it in a feedback loop to the software. The software may be the source of the combination of setpoints related to the test protocols for which certain examples are illustrated in FIGS. **11A**, **11B** and **11C**.

(31) Referring to FIG. **9**, according to an embodiment, each support **4** comprises a pressure plate **94** extending above the active zone **20** of the cell **2** and exceeding a contour **72** of the housing **6**.

(32) Referring to FIG. **3**, the pressure application surface **18** on the active zone **20** (illustrated e.g. in FIG. **9**) defines a cavity **74** having a shape adapted to receive a portion of the pressure plate **94** exceeding the contour **72** of the housing **6** (illustrated e.g. in FIG. **9**). For a pouch-type cell **2**, the cavity **74** may have a generally flat bottom **76**.

(33) Referring back to FIG. **9**, the contour **72** of the housing **6** may have openings **78** through which terminals **80** of the cell **2** project for an electrical connection outside the support **4**, for example with alligator clips (not illustrated). Other types of electrical connections may be used if desired. According to an embodiment, each support **4** may also comprises an insulating contact plate **82** extending against a bottom **84** of the housing **6**. The contact plate **82** may have tabs **86** projecting in the openings **78** of the contour **72** of the housing **6** and forming mechanical supports for the terminals **80** of the cell **2** mounted on the contact plate **82** in the housing **6**, as better illustrated in FIG. **10**. The tabs **86** may also be used to electrically insulate the terminals **80** of a cell **2** with respect to another one. A spacer **88** extending above the cell **2** may be used for a precise positioning of the elements in the support **4** and to even out the pressure on the active zone **20** of the cell **2**. The spacer **88** may have a central portion **90** generally flat and matching with the active zone **20** of the cell **2**. The spacer **88** may be made of a flexible material such as nylon, Teflon, plastic, so as to adapt to surface defects of the cell **2**. Arms **92** projecting in periphery of the central portion **90** come resting against internal faces of the contour **72** of the housing **6** so as to define a positioning of the central portion **90** on the active zone **20** of the cell **2**. The pressure plate **94** extends against the central portion **90** of the spacer **88** and is in contact with the pressure application surface **18** (illustrated e.g. in FIG. **3**) which applies pressure on the active zone **20** of the cell **2** of the clamping arrangement **8** when the jaws **10**, **12** are in the pressure application position. A construction of the support **4** as described above ensures a uniform pressure application on all the surfaces of the cell **2**, and a precise positioning of the cells **2** and the supports **4** in the clamping arrangement **8**. Depending on the practiced cycling mode and the pressure applied to the cell **2**, an elastic arrangement, for example an arrangement of springs **95** as in the illustrated case, introducing a mechanical elasticity in the pressure application axis may be inserted between the pressure application surface **18** (as illustrated in FIG. **2**) and the pressure plate **94**, or between the contact plate **82** and the bottom **84** of the housing **6**, or yet between the housing **6** and the pressure application surface **18**. The elastic arrangement may take another form if desired, for example an elastomeric element or a rubber disk. Such an elastic arrangement provides a play in the vertical displacement of the parts comprised between the jaws **10**, **12** (illustrated in FIG. **2**) when a force is applied by the actuator **22** (illustrated in FIG. **2**). Such a play in the transmission chain of the force may be useful to absorb and allow a certain variation of thickness of a cell **2** during a cycling without requiring an intervention at the level of the actuator **22**, and to prevent damaging the cell **2** in such a case.

(34) Referring again to FIG. **1**, the system according to the invention thus allows testing a response of a battery to a programmed cycling mode and various precise and regulated values of pressure and, if applicable, temperature applied to the cells **2** of the battery representing possible conditions of operation and use in order to determine one or more pressures to be applied to the cells **2** of the battery and other possible physical parameters such as one or more temperatures to obtain an

optimum performance of the battery depending on its chemistry, its architecture and its conditions of operation and use. The system allows applying a range of various and significant pressures, for example, from 0 to 4000 PSI or more to cells **2** having for example an active zone **20** (illustrated in FIG. **9**) of 50 mm×50 mm. The system, with its processing unit **38** and its other components forms a controlled and programmable assembly allowing to maintain a constant pressure and, if applicable, temperature despite a volume variation of the cells in charging and discharging phase which may represent approximately 25% of volume variation, and allowing to vary in real time and without human intervention the pressure or the temperature applied to the cells **2** as a function of a test protocol programmed in the processing unit **38**. The system allows evaluating the effect of the pressure and preferably also of the temperature on the performance of the cells **2** and their resulting behavior, for different chemistries used. It allows cycling several cells **2** at the same time, at different pressure values, and at different temperature values when used in heat chambers or provided with another temperature regulation mean. The system thereby allow determining an effect of the real-time adjustment of the pressure and preferably also of the temperature on the performance of the cells **2** for different conditions of use of a battery (charge, discharge, power demand, conditions prone to a formation of dendrites, use in extreme climatic conditions, number of cycles experienced by the cells **2**, etc.). Knowledge of the optimal operating parameters of the cells **2** of a battery having a particular chemistry allows coding control algorithms specific to the battery.

(35) Referring again to FIG. **2**, in order to allow a uniform distribution of the pressure, the cells **2** are disposed in the supports **4** (according to one embodiment illustrated e.g. in FIG. **9**) so as to precisely position them under the piston **48** applying the force. One or several cells **2** may be tested/cycled at the same time in the system, where the supports **4** are arranged in series. Cells **2** of different thicknesses may be cycled without modification to the system. The system works for different geometries of cells **2**, by adapting the dimensions of the different components. Several clamping arrangements **8** may be simultaneously used to perform tests at various pressures and at various temperatures. The system allows a quick mounting of the cells **2** to be tested.

(36) While embodiments of the invention have been illustrated in the accompanying drawings and described above, it will be evident to those skilled in the art that modifications and additions may be brought to these embodiments without departing from the invention.

Claims

1. A system for cycling battery cells, the system comprising: at least one support defining a housing for receiving one of the battery cells; a clamping arrangement having (i) lower and upper jaws between which said at least one support housing one of the battery cells is insertable, the jaws being mobile with respect to each other in a pressure application axis, and (ii) pressure application surfaces for applying pressure to the cell housed in each support, one of the pressure application surfaces being intended to exert a pressure on an active zone of the cell when the jaws are in pressure application position; an actuator operatively coupled to the clamp arrangement in order to control a clamping of the jaws according to a control signal; a pressure sensor operatively associated to the cell in order to produce a signal indicative of a pressure exerted by the jaws on the cell; a cycling module for cycling the cell, the cycling module having an interface connectable to the cell for charging and discharging the cell according to a programmed cycling mode and measuring a level of charge and discharge of the cell; a control module for controlling the actuator, producing the control signal controlling the clamping of the jaws according to a pressure setting signal; and a processing unit connecting to the cycling module, to the control module of the actuator and to the pressure sensor, the processing unit being configured to program the cycling mode of the cell and transmit the cycling mode to the cycling module, to program a pressure to be applied to the cell and generate the pressure setting signal according to a pressure measured by the

pressure sensor in order to form a feedback loop of the pressure applied to the cell, and to record data representative of the pressure applied to the cell and of the level of charge and discharge of the cell.

2. The system according to claim 1, wherein: said at least one support comprises several supports for receiving as many of said battery cells; and the pressure application surfaces are formed by surfaces of the jaws and surfaces of a pressure transmission plate extending between two of said supports, each pressure transmission plate being freely mobile in the pressure application axis.

3. The system according to claim 2, wherein the clamping arrangement comprises a set of guiding rods extending between the jaws in the pressure application axis, the guiding rods passing through corresponding openings in each pressure transmission plate so that each pressure transmission plate is mobile only in the pressure application axis.

4. The system according to claim 2, wherein each pressure transmission plate has an upper surface defining a positioning cavity in which a lower part of one of said supports is engageable.

5. The system according to claim 2, wherein the clamping arrangement comprises springs extending between the jaws and each pressure transmission plate so that the jaws and each pressure transmission plate are moved away from one another by a distance of at least a height of one support housing a cell when the jaws are in release position.

6. The system according to claim 1, wherein the lower jaw has an upper surface defining a cavity having a shape adapted to receive the pressure sensor so that the pressure sensor measures a pressure applied by the jaws to the cell in said at least one support.

7. The system according to claim 1, wherein the lower jaw forms a base of the clamping arrangement.

8. The system according to claim 1, wherein the upper jaw is formed by a pressure transmission plate freely mobile in the pressure application axis, and the actuator comprises a motor unit, a piston coupled to and moveable by the motor unit in the pressure application axis according to the control signal transmitted to the motor unit, and a support structure for supporting the motor unit at a fixed distance from the lower jaw, the piston having an end pressing against the pressure transmission plate forming the upper jaw when the jaws are in said pressure application position.

9. The system according to claim 1, wherein the pressure application surfaces are substantially flat and extend perpendicular to the pressure application axis.

10. The system according to claim 1, wherein the processing unit comprises a processor coupled to a memory, the memory storing instructions executable by the processor and operating parameters of the system received as input, the operating parameters comprising a test protocol used by the instructions executed by the processor and defining the pressure to be applied.

11. The system according to claim 10, wherein the test protocol defines a series of pressures to be applied and maintained during a time period, the pressure setting signal transmitted to the control module of the actuator being regulated by the processing unit in compensation to a volume variation of the cell during an execution of the cycling mode.

12. The system according to claim 11, wherein the instructions comprise instructions for dynamically varying the series of pressures to be applied and maintained as a function of cycles or a rate of charge and discharge of the cell.

13. The system according to claim 1, further comprising a temperature sensor operatively associated to the cell in order to produce a signal indicative of a temperature of the cell, the processing unit connecting to the temperature sensor and being configured to record data representative of the temperature of the cell according to the signal indicative of the temperature of the cell, the pressure setting signal being adjusted according to the temperature measured by the temperature sensor.

14. The system according to claim 1, further comprising a temperature sensor operatively associated to a cell in order to produce a signal indicative of a temperature of the cell, and a temperature adjustment module acting on a heat-exchange element flowable around the cell as a

function of the signal indicative of a temperature produced by the temperature sensor and a temperature setpoint generated by the processing unit.

15. The system according to claim 1, wherein said at least one support comprises a pressure plate extending above the active zone of the cell and exceeding a contour of the housing, and the pressure application surface on the active zone defines a cavity having a shape adapted to receive a portion of the pressure plate exceeding the contour of the housing.

16. The system according to claim 15, wherein the cavity has a substantially flat bottom.

17. The system according to claim 15, wherein said at least one support further comprises: an insulating contact plate extending against a bottom of the housing, the contact plate having tabs projecting in openings of the contour of the housing and forming mechanical supports for terminals of the cell mounted on the contact plate in the housing which project through the openings; and a spacer extending above the cell, the spacer having a central portion substantially flat and matching with the active zone of the cell, and arms projecting in periphery of the central portion and pressing against the contour of the housing in order to define a positioning of the central portion on the active zone of the cell, the pressure plate extending against the central portion of the spacer, the pressure plate being in contact with the pressure application surface on the active zone of the cell of the clamping arrangement when the jaws are in the pressure application position.

18. The system according to claim 1, wherein the housing has a contour having openings through which terminals of the cell project for a connection external to the support.

19. The system according to claim 1, further comprising an elastic arrangement inserted between the pressure application surfaces in order to introduce a mechanical elasticity in the pressure application axis.
